

# Arihant Jain

Senior Backend Engineer | Distributed Systems | 5+ Years

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## Skills

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| Languages                 | Javascript(ES6+), <b>Python</b> , SQL                                  |
| Backend Frameworks & APIs | Node.js, <b>FastAPI</b> , REST APIs, Pydantic                          |
| Data & Messaging          | Redis, MongoDB, Kafka, RabbitMQ, Celery, Amazon SQS                    |
| Cloud & Delivery          | AWS Lambda, Fargate/ECS, S3, EC2, CloudFormation, Docker, GitLab CI/CD |
| Testing & Observability   | Pytest, Playwright, Selenium, AWS CloudWatch, VAPT workflows           |

## Experience

### Senior Software Engineer

Apr 2024 – Present

Karza Technologies Pvt Ltd. (A Perfios Company)

New Delhi, IN (Hybrid)

- Architected an identity routing engine that reduced third-party vendor dependency by **55%** and delivered **INR 6.5 Cr+ (\$780K+) in annual savings**.
- Built a Redis-backed caching layer with controlled fallback logic, reducing redundant calls and generating **INR 22L/month (\$26K)** in savings.
- Revamped a legacy rendering engine into a Playwright worker cluster, cutting latency by **85% (40s → 6s)** and improving node uptime from **60% to 98%**.
- Owned disaster recovery for Mumbai/Hyderabad and shipped infrastructure for **16+ international APIs** across Kenya and UAE, supporting **5M+ daily requests at 99.9% SLA**.
- Migrated **260 production APIs** to Python 3.12 with **99% containerization**, standardizing CI/CD and enabling safer parallel regional deployments.
- Spearheaded pilot-crawl rollout across **20 crawler APIs** using manual and AI-assisted migration workflows; mentored new hires, conducted **50+ in-depth code reviews**, and caught a critical fallback-logic defect before release.

### Software Engineer I

Oct 2022 – Apr 2024

Karza Technologies Pvt Ltd. (A Perfios Company)

Mumbai, IN (Remote)

- Built **30+ REST APIs** from scratch and maintained **100+ APIs** that aggregated real-time data from **300+ public registries**, supporting **1M+ daily transactions**.
- Re-engineered high-traffic API response paths, cutting **P99 latency by 30%** and improving reliability for client-facing data enrichment workflows.
- Migrated distributed Celery workloads to AWS Fargate, saving **INR 20L/month (\$24K)** and cutting API latency by **1.2s**.
- Automated billing and usage-tracking workflows using scripts and cron jobs, improving billing accuracy by **25%** and reducing processing time by **30%**.
- Resolved database query bottlenecks to boost throughput by **20%**, eliminating request throttling during peak load.

### Software Engineer

Mar 2021 – Sep 2022

InTimeTec

Jaipur, IN (Onsite)

- Drove a **0-to-1 migration** from a legacy .NET desktop application to an Angular web platform across **9 modules**, tripling operational throughput.
- Delivered the project end-to-end across development, deployment, client communication, and release support while coordinating a **4-engineer pod**.
- Reduced deployment cycles by **44%** and increased test coverage by **54%** through structured CI/CD pipelines and Karma/Jasmine suites.

## Projects

### Distributed Quota-Metering Engine | FastAPI, Redis Lua, MongoDB, Docker Compose

GitHub

- Developed a prepaid org/feature-level quota engine where a single Redis Lua script handles idempotency lookup, quota checks, deduction, and denial recording atomically.
- Designed **all-or-nothing batch semantics**, **request-tied idempotent refunds**, and **calendar-month UTC resets** using period-based Redis keys without cron-based reset jobs.
- Validated correctness under contention with **10,000 simulated consume requests**, capped exactly at configured quota with **0 over-served units**; same-key retries were charged only once.

## Education

### Swami Keshvanand Institute of Technology

Aug 2017 – Jun 2021

B.Tech, Information Technology

Jaipur, IN

- Secured **All India Rank 91** in Round 1 of TCS CodeVita Season 9 (2020).